

RUDRA CHAVDA

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EDUCATION

University of California, Santa Barbara

Expected Spring 2028

Bachelor of Science, Computer Engineering

University of California, Santa Cruz

2024 – 2026

Bachelor of Science, Computer Engineering

GPA: 3.95

– **Honors:** 6x Dean's List

Coursework: Systems programming, computer architecture, operating systems, data structures and algorithms, and software engineering.

TECHNICAL SKILLS

Languages Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly

Frameworks Vue.js, React, Next.js, Node.js, Spring Boot, Py4Web, Tailwind CSS, Angular

Tools Git, Github, MySQL, Flyway, Microsoft Graph API, JUnit, Docker, Nautilus (Cluster), CARLA, GDAL

Concepts LLM Agents, Function Calling, OAuth 2.0, Reinforcement Learning, System Design, Data Structures, Algorithms

EXPERIENCE

Ascend Cargo

Jun 2026 – Sep 2026

AI/ML Intern

- Built an **LLM agent framework (Java 23, Spring Boot 3.4, OpenAI function calling)** with annotation-driven tool registration, automatic JSON-Schema generation, and LLM-based intent routing that scopes which tools each request may reach.
- Shipped natural-language **create/update for freight shipments and orders**, replacing multi-screen data entry; diagnosed and fixed a synchronous automation call that blocked shipment creation for **12 minutes, cutting it to under 30 seconds**.
- Designed a **two-gate authorization model** enforcing a per-user AI access ceiling (**NONE/READ/WRITE**) alongside existing app privileges, guaranteeing the agent can never exceed the permissions of the human it acts for.
- Delivered an **inbound email channel over the Microsoft Graph API (OAuth 2.0 client credentials)** letting verified staff create and modify shipments by emailing a shared mailbox, with sender verification resolved per-user from the database.
- Hardened agent behavior after live-demo failures: replaced silent fallbacks with **refuse-and-instruct** error contracts that tell the model how to recover, eliminating a class of confidently-wrong records.
- Authored **122 unit tests (JUnit 5, Mockito)** plus a CLI harness that drives the chat API and verifies results against the database, and integrated 3 months of upstream divergence across 38 conflicting files.

AIEA Lab (UCSC)

Dec 2025 – Apr 2026

Undergraduate Researcher

- Wrote and benchmarked Reinforcement Learning (RL) algorithms such as **DQN, DDQN, A3C**, and **Stable Baselines** in the Car Racing environment to evaluate autonomous vehicle safety as part of the Robustifying AV's project.
- Deploying the **CARLA** simulator in a headless environment on the **Nautilus** hyper-cluster (Kubernetes).
- Analyzed edge-case failures by visualizing agent behaviors and reward functions post-training.

CA-Futures (UCSC EEB & Baskin Engineering)

May 2025 – Sep 2025

Software Developer

- Developed a full-stack conservation modeling application using **Vue.js** (frontend) and **Py4Web** (backend).
- Integrated geospatial analysis tools (**GDAL, Ecoscape**) to track species connectivity and visualize ecological data.
- Optimized simulation parameters to process large-scale biological datasets for the Ecology Lab.

Opennote (YC S25)

Jul 2025 – Feb 2026

Campus Growth Lead

- Co-organized UCSC Cruzhacks in collaboration with Opennote where we gave out \$1500 worth of prizes.
- Organized workshops talking about user-agent API flow in typescript.
- Spearheaded campus growth strategy, increasing user engagement by 30% and acquiring 50+ active users.

Front-End Engineering Intern

- Architected a secure, interactive web application using **React.js** and **Tailwind CSS**.
- used **Apache Echarts** to visualize client side data
- Designed a responsive front-end framework to improve data visualization for client dashboards.

PROJECTS

Better Notes | *TypeScript, Tailwind CSS, Vercel*

Spring 2025

- Engineered a full-stack note-taking platform with a custom markdown rendering engine for student study workflows.
- Secured 100+ active users within 3 months of deployment to Vercel.
- Live Demo: betternotes-two.vercel.app

Div Explorer Project | *Machine Learning, Python*

Fall 2025

- Developed actionable data labeling pipelines to detect duplicate discoveries and high-overlap patterns.
- Implemented categorization metrics (interpretability, coverage, anomalies) to improve ML model fairness.

Personal Portfolio | *Next.js, Three.js, Motion*

Winter 2024

- Built a high-performance portfolio site featuring 3D interactive elements using Three.js and Motion animations.

LEADERSHIP

Eagle Scout Project

Summer 2024

Project Lead, Troop 537

- Directed a team of 20+ Scouts in the construction of redwood structures for Sycamore Grove Park (Livermore).